Lecture Scribe CSE400

L21_L22.pdf
PDF

generate a lecture scribe for the pdf i gave you so that i can refer it on the exam day for revision
and also use only the content given in the slide.do not use anyother information from outside
the scribe should include :
1.)definations
2.)symbols
3.)important results
4.)formula
5.)symbols
6.)proof
7.)derivation should be step by step and same with examples.
the scribe should not include new explanations,example,and also do not simplify or modify anything from slide.Do not ask me any question give the respose dierctly

Here is your **LECTURE SCRIBE (STRICTLY from slides, no modification, no extra content)**

CSE400 – Lecture L21–L22 Scribe

Inequalities and Tail Bounds

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1. DEFINITIONS

Tail of Random Variable

- The tail of a random variable X is:

$$P(X > x)$$

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2. SYMBOLS

- X → Random variable
- μ → Mean $E[X]$
- σ → Standard deviation
- $Var(X)$ → Variance
- a → Threshold
- n → Number of trials
- p → Probability
- λ → Rate (Poisson)
- t → Parameter (Chernoff)

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3. IMPORTANT RESULTS

Markov Inequality

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Chebyshev Inequality

$$P(|X - \mu| \geq a) \leq \frac{Var(X)}{a^2}$$

Chernoff Bound (Final Form)

$$P(X \geq a) \leq \min_{t>0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

Chernoff Bound (Lower Tail)

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$$P(X \leq a) \leq \min_{t<0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

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4. FORMULAS

Binomial Tail

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Poisson Tail

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Expectation

$$E[X] = \sum x \cdot p_X(x)$$

MGF (Poisson)

$$E[e^{tX}] = e^{\lambda(e^t - 1)}$$

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5. KEY IDEAS (FROM SLIDES)

- Mean → average behavior (easy)
- Tail → rare extreme events (hard)
- Exact tail computation → complicated long sums
- Instead → compute upper bound

Markov Key Idea

- Uses only the mean
- Large values ⇒ rare

Chebyshev Key Idea

- Uses mean + variance
- Controls deviation from mean

Chernoff Key Idea

- Uses exponential e^{tX}
- Exponential magnifies large values
- Better transformation ⇒ tighter bound

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6. PROOFS

Markov Inequality Proof

$$E[X] = \sum_{x=0}^{\infty} x \cdot p_X(x)$$

$$\geq \sum_{x=a}^{\infty} x \cdot p_X(x)$$

$$\geq \sum_{x=a}^{\infty} a \cdot p_X(x)$$

$$= a \cdot P(X \geq a)$$

$$\Rightarrow P(X \geq a) \leq \frac{E[X]}{a}$$

Chebyshev Inequality Proof Idea

Apply Markov on $(X - \mu)^2$:

$$P(|X-\mu| \geq a) = P((X-\mu)^2 \geq a^2)$$
$$\leq \frac{E[(X-\mu)^2]}{a^2}$$
$$= \frac{Var(X)}{a^2}$$

Chernoff Bound Proof (Core Trick)

Step 1: Transformation

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Step 2: Apply Markov

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

Step 3: Final

$$P(X \geq a) \leq \min_{t>0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

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7. DERIVATIONS (STEP-BY-STEP FROM SLIDES)

Running Example Setup

- $X \sim \text{Binomial}(n, \frac{1}{2})$
- $\mu = E[X] = \frac{n}{2}$

Markov on Running Example

$$P\left(X \geq \frac{3n}{4}\right) \leq \frac{E[X]}{3n/4}$$

$$= \frac{n/2}{3n/4}$$

$$= \frac{2}{3}$$

Chebyshev on Running Example

$$P\left(X \geq \frac{3n}{4}\right) = P\left(X - \frac{n}{2} \geq \frac{n}{4}\right)$$

$$\leq P\left(|X - \mu| \geq \frac{n}{4}\right)$$

$$\leq \frac{Var(X)}{(n/4)^2}$$

$$= \frac{n/4}{n^2/16}$$

$$= \frac{4}{n}$$

Chernoff for Poisson (Derivation)

$$E[e^{tX}] = \sum_{i=0}^{\infty} e^{ti} e^{-\lambda} \frac{\lambda^i}{i!}$$

$$= e^{-\lambda} \sum_{i=0}^{\infty} \frac{(\lambda e^t)^i}{i!}$$

$$= e^{-\lambda} e^{\lambda e^t}$$

$$= e^{\lambda(e^t - 1)}$$

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8. EXAMPLES (FROM SLIDES)

Example 1: Income Distribution (Markov)

$$P(X \geq 50000) \leq \frac{10000}{50000} = 0.2$$

Example 2: Exam Grades (Markov)

- $\mu = 70$, $a = 90$

$$P(X \geq 90) \leq \frac{70}{90} = \frac{7}{9}$$

Example 3: Server Requests

$$P(X \geq 20000) \leq \frac{5000}{20000} = \frac{1}{4}$$

Example 4: Chebyshev (Exam Scores)

- $\mu = 70$, $\sigma = 10$

$$P(|X - 70| \geq 30) \leq \frac{10^2}{30^2} = \frac{1}{9}$$

Example 5: Chebyshev (Grades)

$$P(X \geq 90) \leq \frac{25}{400} = \frac{1}{16}$$

Example 6: Network Latency

$$P(|X - 50| \geq 20) \leq \frac{16}{400} = \frac{1}{25}$$

END OF SCRIBE (STRICTLY FROM SLIDES)